

Leukemia Detection using Computation Intelligence Techniques

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Abstract— The discipline of molecular biology is the focus of the application of statistics and computer science that is known as bioinformatics. The term "bioinformatics" refers to the study of how mathematics, statistics, and information technology might be used to the study of biological sciences. The bioinformatics of microarrays is the field's response to the challenge posed by this definition. Genes are units of information that are responsible for encoding the genetic material found in all creatures that are alive. The term "genome" refers to the whole collection of genes that are included inside an organism. For the purpose of collecting, analyzing, storing, and exchanging genetic information with other researchers, the design of studies using microarrays is required. Because they are an organized collection of thousands of distinct Deoxyribonucleic Acid (DNA) sequences that can be used to measure variations in both DNA and Ribonucleic Acid (RNA), micro-arrays have acquired a special significance in the field of bioinformatics. This is due to the fact that micro-arrays can be used to measure both DNA and RNA. In recent years, machine learning and artificial intelligence have seen fast advancements, which has led to the methodology being one of the most widely used methods. Major corporations and educational institutions have begun investing in research in the healthcare industry with the goal of improving the accuracy with which diseases can be predicted as a result of its growing popularity and the powerful techniques it employs in pattern recognition and classification. However, while putting these strategies into practice, there are a lot of obstacles to overcome. The absence of a huge data collection including medical photographs is one of the most significant challenges that must be overcome. Because they make it possible to do quantitative research on thousands of genes all at once using a single sample of cells, microarrays in particular are being hailed as a major technological advancement in the field of biology. In the first model of an artificial neural network, two different classification algorithms—feed forward backpropagation and cascade correlation feedforward—were compared with different sets of neurons using two different training algorithms—Levenberg Marquardt (lm) and Resilient backpropagation (rp), in order to differentiate between benign and malignant patients. According to the findings, cascade correlation with the train (rp) produced superior output outcomes when compared to feedforward back-propagation with the train (lm).

Keywords—Cancer detection, Leukemia Detection, Levenberg Marquardt, Resilient backpropagation.

I. INTRODUCTION

Imaging in medicine refers to any method or procedure that examines the components, organs, and tissues of the body

in order to make a diagnosis of illness. Image processing and computational intelligence approaches provide digital medical pictures utilizing which various phenomena may be investigated and explored with the assistance of computers [1]. The discipline of medicine has benefited considerably from the exponential progress that has been made in research and development within the field of image analysis. When it comes to the diagnosis and evaluation of a wide variety of disorders, such as those affecting the breast, lungs, brain, abdomen, and blood, among other organs and systems, medical pictures are widely regarded as an essential diagnostic tool.

Images of a medical condition stored in digital format allow for more in-depth investigation, which ultimately leads to a more accurate diagnosis. These kinds of photos may also be put to use for the purposes of study and instruction. This study makes use of microscopic pictures of peripheral blood [2], which are represented as digital representations in the medical field. One of the frequent diagnostic procedures that are performed in clinical routine practice is the study of blood components and the changes that occur in them.

The cancerous growth known as leukemia affects both the blood and the bone marrow of the patient. The creation of abnormal white blood cells inside the human body is the root cause of this condition. Blood is a mixture of millions of different cell types that exists in liquid form. Red blood cells (RBCs), white blood cells (WBCs), platelets, and plasma are all included in its make-up. It is also known as the different components of blood cells. Only one percent of the blood in the human body is made up of white blood cells, which are responsible for fighting illness. These WBCs are the elements that are most severely impacted by cancer. Forty to forty-five percent of red blood cells (RBCs) are responsible for transporting oxygen throughout the body [3]. Platelets are less substantial cells that play an important role in the regulation of blood loss. Plasma is the name given to the liquid component of the blood. It assists in the delivery of water and nutrients to the tissues of the body. According to information found on the website <https://www.oneblood.org>, the term "Bone Marrow" refers to the location where all of the blood components are manufactured. Because of leukemia, the bone marrow produces a high number of abnormal white blood cells that have not yet reached their full developmental potential [4]. The microscopic pictures shown in Fig. 1 compare healthy blood with blood that has been contaminated with leukemia.

II. LEUKEMIA CLASSIFICATION

The STRSFFQR approach, which was recommended, has delivered the lowest RMSE value, which is defined as an average that is lower than 0.25 across all classification methods. This definition was supplied by the STRSFFQR method. A technique known as the Improved Dominance Soft Set-based Decision Rules [5] with Pruning (IDSSDRP) algorithm was used in the images of acute lymphoblastic leukemia in order to discern between the blast (leukemia) cells and the non-blast (normal) cells. On the basis of the results of the experiments, it is possible to draw the following conclusions regarding the overall classification accuracy of the proposed IDSSDRP: 98.08 percent for the GLCM 0 dataset; 97.12 percent for the GLCM 45 dataset; 99.04 percent for the GLCM 90 dataset; 97.60 percent for the GLCM 135 dataset; and 95.67 percent for the shape dataset and the color dataset, respectively [6]. You can find the ROC curve that was generated by the IDSSDRP approach at the top left edge of the ROC graph, which becomes more significant as it proceeds down. This demonstrates that the technology that is now under discussion is able to properly distinguish blast cells from other types of cells that are not blast cells.[18-19]

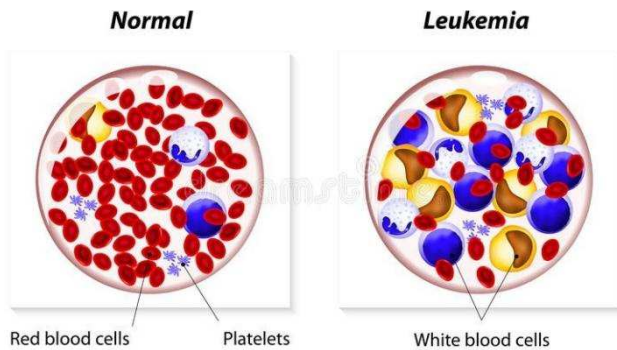


Fig. 1. Microscopic View of Leukemia

III. LITERATURE SURVEY

In their analysis of prostate MRI images, [7] used segmentation. Stick filtering and contrast stretching were used in the preprocessing stage to eliminate the noise that was present in a picture. The image of the prostate gland was extracted from the axial MRI slices and segmented using a fuzzy C-means (FCM) clustering technique for the purpose of this study. Using this method, the region of interest (ROI) in the picture of the prostate was automatically segmented.

A unique fuzzy clustering approach for the segmentation of dental X-Ray pictures was developed by [8]. This algorithm is based on the neuromorphic orthogonal matrices. Within the context of this method, the picture data was converted into a neuromorphic set, and the cutting matrix of an input image was identified. For the purpose of this study, dental datasets were obtained from the Hanoi Medical University Hospital in Vietnam in order to conduct an analysis of the effectiveness of the suggested technique.

In [9] presented a method for magnetic resonance imaging (MRI) brain tumor picture segmentation that makes use of the Soft Fuzzy Rough C-Means (SFRM) clustering algorithm. This approach for clustering was developed by combining ideas from soft set theory, fuzzy set theory, and rough set theory. Instead of using the crisp values, the approximations of the cluster were computed using the values after they had been fuzzified. The histogram of the picture was used in this

method in order to determine the centroids of the image. The SFRM clustering method has superior results than the other clustering algorithms that were already available, namely the K-Means clustering algorithm, the Rough K-Means clustering algorithm, and the Fuzzy C-Means clustering algorithm.

In [10] came up with the idea of an unsupervised segmentation of lymphoma pictures taken from histology slides. Histogram equalization and the Gaussian filter were used, respectively, in this investigation to improve the image's overall quality. In order to separate the nuclei of normal cells from those of malignant cells in histology pictures, a hybrid Genetic Algorithm (GA) and fuzzy 3-partition entropy approach were used. Comparisons were made between this approach and other bio-inspired algorithms already in existence, such as Artificial Bees Colony (ABC), Cuckoo Search (CS), Differential Evaluation (DE), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Wind Driven Optimization (WDO). The suggested approach produced quantitative findings that were superior overall.

In [11] introduced the Fuzzy Spectral Clustering (FSC) technique to segment pictures of wound areas. The ulcer border in the picture was marked using this method after it was determined to be the most effective. Validation of the suggested technique was accomplished by applying it to ground-truth photographs annotated by two knowledgeable dermatologists. When it came to picture segmentation, the FSC algorithm generated superior results.

In [12] created a diagnosis approach for pictures of acute myeloid leukemia (AML). The counting of blast cells in the blood samples was the only factor that mattered in this methodology. K-Means clustering was used in order to carry out the process of blast cell segmentation. The Hidden-Markov Random Field algorithm was used to isolate the nucleus and cytoplasm from each of the pictures (HMRF).

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The K-Means clustering technique was used for the purpose of picture segmentation by [16]. The histogram of a picture was used to determine the best possible number of clusters to use. During the course of this research, the MATLAB media picture dataset was studied, and the

suggested approach generated satisfactory results when applied to image segmentation issues.

Researchers from [17] developed a Computer Aided Diagnosis (CAD) system with the goal of distinguishing blast cells from healthy lymphocytes. When compared to unique characteristics, the texture- and shape-based features used in this detection functioned as superior indications. This detection was based on those features.

Patients are able to continue receiving therapy in their native environments because to RTHMS and MHMS, which are more trustworthy and cost-effective than conventional care. Those health monitoring systems that patients wear, such as real-time monitoring systems (RHMS) and portable monitoring devices, are examples of what are referred to as weakening health monitoring systems (WDHMS). Since 2010, a number of researchers have made use of wearable sensors such as accelerometers and gyroscopes to determine the closeness of a moving item and the direction it is travelling. This list of twenty things includes both wearable technology and biosensors in some capacity. When it comes to matters of healthcare, a cautious approach is adopted, and as a direct result of this, fashionable devices that may be worn on the body are included in this category. In light of previous investigations, a new and better healthcare monitoring system that makes use of the most recent and cutting-edge technologies has been presented.

Several different computing strategies that have been used in the research literature to analyze photos of tumors were addressed. The presentation also included a summary of current research activities in leukemia image analysis, including preprocessing, nucleus segmentation, feature extraction, feature selection, and classification. It was provided in detail how a summary of the imaging modalities, types of illnesses and pictures, segmentation approach, feature extraction and selection, and classification methods were employed.

IV. MATERIALS AND METHODOLOGY

Both the ALL-IDB1 and ALL-IDB2 databases include digitized photographs of microscopic examinations of blood samples. These databases were created specifically for the purpose of researching acute lymphoblastic leukemia (ALL). It is put to use to a significant extent for the aim of evaluating and comparing a variety of different segmentation and classification methods. These data, which were acquired by downloading them from the website situated at <https://homes.di.unimi.it/scotti/all/>, are the property of the Fabio Scotti Department of Informatics at the University of Milan, Italy. You may find the website at <https://homes.di.unimi.it/scotti/all/>.

Each "Imax Y.jpg" image that is included in the ALL-DB1 dataset is accompanied with a "Imax Y.xyc" text file that contains an equivalent classification of the image. The spot in the text file that corresponds to the centroid coordinates of each candidate's lymphoblast image is where those centroid coordinates are saved. A skilled haematologist will assign the classification of ALL or the centroid of lymphoblast to each image before presenting it to the patient. The ALL-DB1 collection includes 108 photos, each of which has a resolution of 2592 by 1944 pixels and is formed of 108 individual images. The ALL-IDB2 dataset is a collection of photographs of normal and leukaemia cells that have been cropped to focus on a particular area of interest. The ALL-IDB1 dataset includes this collection in its aggregation. The

ALL-DB2 collection is comprised of 260 photographs, each of which has a resolution of 257 pixels on each of its two rows of 257 pixels. The data that were acquired from these photographs are what will be used to determine how successful the classification algorithms are. We collected a total of 368 photographs, comprising 175 images depicting normal cells and 193 images illustrating abnormal cells. Fig. 1 displays an example of aberrant and normal photographs that were generated from the ALL-DB1 (a, b) and ALL-DB2 (c, d) datasets, respectively.

The procedure known as segmentation may be used to reduce the complexity of the representation of a picture, which ultimately results in an image with a deeper level of meaning. The process of analyzing images is simplified as a result of this. The process of segmentation is a key step in many various aspects of image processing, including medical image analysis, the identification of objects, the detection of tumors, satellite imagery analysis, and many more of these and other types of analyses. "Segmenting an image of a leukemia nucleus" is one of the diagnostic procedures for leukemia that is known to be one of the most challenging and time-consuming procedures. Many distinct approaches to market segmentation have been developed and used over the course of the last few decades. One of the more recent advancements that has been made in the area of medical image processing is the use of soft computing. Throughout the whole of this proposed work, the Covering-based Rough K-Means (CRKM) clustering technique and the Histogram method are used as two of the primary kinds of segmentation methodologies that are applied. A method of clustering called Soft Covering-based Rough K-Means (HSCRKM) is utilized in order to segment the image of the leukemia nucleus. This method was developed by the University of Michigan.

In the first technique, covering-based approximation is utilized rather than rough approximation. This is because covering-based approximation is more accurate. This is due to the fact that an estimate based on coverage is more accurate. In the second method, the peak values of an image's histogram are found and used for the initialization of the number of clusters. This is done using the methodology described above. The preceding approach, which was explained above, is then followed by this method. As a result of taking this step, the cluster numbers will not be produced randomly at the beginning of the process.

The HSCRKM method takes use of the advantages provided by both the soft covering-based rough set and the Rough K-Means clustering algorithm in order to properly segment the image of the nucleus. The soft covering rough approximation is used whenever it is necessary to determine the lowest and maximum values for the approximation.

The segmented image of the leukemia nucleus is used to extract a variety of properties, such as GLCM-0, GLCM-45, GLCM-90, and GLCM-135 in addition to shape- and color-based characteristics. These features are utilized to help diagnose and treat leukemia. These days, quite a few distinct machine learning algorithms are used in order to make forecasts regarding the seriousness of an ailment. Using cutting-edge machine learning prediction algorithms, dangerous leukemia cells are separated from non-cancerous leukemia cells in the classification process. Fig 2: A Block Diagram of the Work Being Proposed

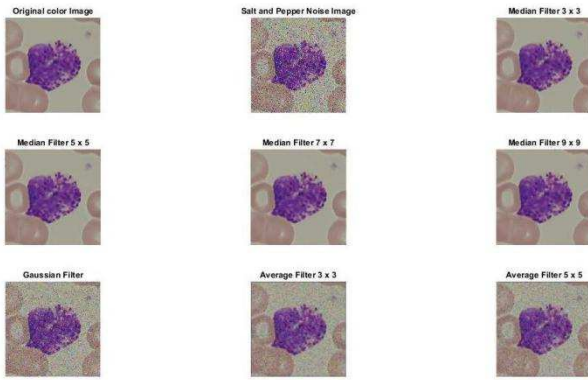


Fig. 2. Filtering stage output

The K-Means clustering approach is one of the hierarchical clustering methods that is used the most often. Within the confines of this method, every single data point is split up into a collection of K different groups. It's possible to divide it up into a couple of distinct phases. The first step is to calculate the centroids, and the second step is to place each data point in the cluster that contains the centroid that is geographically located in the closest proximity to that particular data point. Centroids are calculated in the first step, and data points are placed in clusters in the second step.

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The use of the Euclidean distance is one of the most prevalent tactics that is used in the process of finding the distance that exists between two data points. The following stage, which occurs after the data have been clustered, is to compute a new centroid for each of the clusters. Once again, every data point is positioned inside the appropriate clusters that are the most spread out from one another. This process is done as many times as necessary until all of the data points have been categorized into their respective groups. K-means clustering is an iterative clustering approach that aims to minimize the intra cluster distance that separates each data point from its corresponding centroid. K-means clustering is also frequently known as clustering by means of k-means.

A. Pseudo code of the classifier

The K-means clustering approach has a variety of advantages, the most significant of which are as follows:

(1) It is straightforward to implement; (2) It can analyze massive datasets rapidly and accurately. K-Means clustering does, however, have a few downsides, some of which include the following: (1) Since the beginning centroid is chosen at random, it is probable that the results will be different for each initial centroid. This is because each initial centroid was analyzed separately. As a result, having the proper level of expertise is crucial in order to choose a suitable starting centroid. (2) One more challenging task involves figuring out the total number of clusters, which is represented by the letter

K. The Acute Lymphoblastic Leukemia Image Datasets, more especially ALL-IDB1 and ALL-IDB2, were used for the aim of this inquiry. We used a total of 368 pictures for the purpose of our experimental inquiry, with 175 of them depicting normal cells and 193 of them exhibiting leukemia cells. After the preprocessing stage was finished, the CRKM and HSCRKM clustering algorithms were used to the photos of the leukemia nucleus in order to correctly segment them. K-Means, FCM, and RKM are the three modern clustering algorithms that we use to compare the performance of the recommended methods to. This allows us to assess the efficacy of the aforementioned methods.

The results of applying the recommended CRKM clustering-based segmentation algorithms to the photographs Im148 0, Im212 0, Im100 1, and Im072 1, the segmentation was successful. These photographs are labelled as follows: 148 0, 212 0, 100 1, and 072 1. The initial photo, the image after clustering, and the pictures for cluster1, cluster2, and cluster3 are shown in Fig. 2 correspondingly. The HSCRKM method begins by locating the histogram values that are at their highest point, and then assigns a cluster number depending on the total number of peak values. The value of K is subject to vary at each repetition. In Figure 4 the first column shows the original picture, the second column displays the histogram of the image, which helps to determine the number of clusters (K), the third column represents the clustered image, and the final column shows the image of the nucleus after it has been removed. It has been found that the K value should be as low as possible for the segmentation outcome to be improved. The processing time may be cut down with careful monitoring of the K value.

According on the empirical findings, the performance of a variety of clustering-based segmentation methods is evaluated and compared to that of contemporary machine learning techniques. The extracted feature set was input into the Machine Learning (ML) prediction algorithms, and the results were used to identify the segments of the picture depending on whether or not they included tumor or non-tumor leukemia cells. From the segmented picture of the leukemia nucleus, many different kinds of characteristics, such as GLCM-0, GLCM-45, GLCM-90, and GLCM-135, as well as shape- and color-based features, were retrieved. In this experiment, five different Machine Learning (ML) techniques, namely Logistic Regression (LR), Naive Bayes (NB), Support Vector Machine (SVM), K-Nearest Neighborhood (KNN), and Neural Network (NN), were used in order to assess the effectiveness of the clustering algorithms.

Using a variety of assessment measures, such as accuracy (A), precision (P), recall (R), F1 measure, and mean absolute error (MAE), the performance of the machine learning prediction algorithms was evaluated and examined. The effectiveness of the segmentation algorithms was evaluated using a variety of machine learning prediction techniques, which were taken into consideration.

The results of the experiments demonstrate that the CRKM and HSCRKM clustering algorithms, which were suggested, perform much better than the other methods. It is believed that logistic regression and neural networks perform well when compared to other prediction algorithms and also produce the highest classification accuracy, which is equal to 93 percent. This conclusion was reached after conducting a more in-depth analysis of the overall performance of the proposed methods.

In addition to this, it has been shown that the Naive Bayes approach has the lowest accuracy rate of classification, which is equal to 58 percent.

The overall accuracy of the clustering methods K-Means, FCM, RKM, CRKM, and HSCRKM. All of the machine learning techniques provide a lesser level of prediction accuracy in compared to K-means clustering, which may be expressed as a percentage less than 80. It should be mentioned that with regard to the RKM and FCM algorithms, some ML approaches (namely, Logistic Regression) give rise to prediction accuracy that is higher than 80%. When it comes to the CRKM and HSCRKM clustering algorithms, the majority of ML approaches (with the exception of Naive Bayes) obtain an accuracy of prediction that is more than 80 percent. In addition to this, it can be deduced that the clustering algorithms CRKM and HSCRKM that were presented are superior in their ability to divide the nucleus picture. There is a good chance that the extracted characteristics (which are based on the segments) improve the prediction accuracy. In order to evaluate the results of the experiment, we have experimentally determined that the optimum accuracy is more than 80 percent.

A Covering-based Rough K-Means clustering (CRKM) technique and a hybrid Histogram were both proposed in this study. In order to segment the picture of the leukemia nucleus, a Soft Covering-based Rough K-Means clustering approach was used. The CRKM algorithm has a number of advantages, including the following: (1) The lower and upper approximation of the clusters may be found by using a covering-based rough set approximation. (2) The picture of leukemia cells may be segmented very successfully using this approach.

The fact that the HSCRKM clustering method uses the soft covering rough approximation rather than the regular rough approximation is the primary benefit of using this approach. It is an innovative kind of soft rough set that is able to cope with uncertainties in an efficient manner. In this procedure, the cluster numbers are first given their default values by the histogram. The following criteria are used in the evaluation of the results: (1) The effectiveness of the suggested clustering methods is evaluated in comparison to the performance of well-known and widely used clustering algorithms. These algorithms include K-means clustering, FCM clustering, and Rough K-Means clustering. (2) Using conventional assessment measures, the most advanced prediction approaches in machine learning (ML) are compared with one another.

Based on the findings of the experiments, it is hypothesized that the CRKM and HSCRKM clustering algorithms provide over an 80 percent accuracy rate in their predictions. It should also be mentioned that logistic regression and neural networks both produce an accuracy rate that is often more than 90 percent and perform better than other techniques of prediction. The HSCRKM approach has the drawback that the number of peak values in the histogram increases when it is used to pictures with many colors, such as satellite photos, agricultural images, photographs, and so on. As a result, the amount of time required for processing these images also increases.

In order to pick the most useful characteristics from the extracted feature set, we used the Soft Set-based Unsupervised Quick Redact (SSUSQR) algorithm, the Modified Dominance Soft Set-based feature selection Algorithm (MDSSA), and the Supervised Tolerance Rough Set Firefly-based Quick Redact (STRSFFQR) algorithm. Table 1 presents the comparison of existing works with proposed work.

TABLE. I. COMPARISON OF EXISTING WORKS WITH PROPOSED WORK

Predicti on Metrics	Decisio n Tree [10]	J48 [11]	JRip [12]	LMT [13]	Rando m Tree	Propose d
Accuracy (%)	81.25	81.73	79.81	81.73	80.29	95.67
Sensitivity (%)	96.58	95.89	92.47	95.21	92.47	97.26
Precision (%)	80.57	81.40	81.33	81.76	81.82	96.60
Error Rate	0.19	0.18	0.20	0.18	0.20	0.04
MCC	0.52	0.54	0.48	0.54	0.50	0.90
F1-score (%)	87.85	88.05	86.54	87.97	86.82	96.93
G-mean (%)	88.21	88.35	86.72	88.23	86.98	96.93
Lift value	1.15	1.16	1.16	1.16	1.17	1.38
Youden's index	0.42	0.44	0.42	0.45	0.44	0.89
BCR (%)	70.87	72.14	71.23	72.60	72.04	94.60
BER	0.29	0.28	0.29	0.27	0.28	0.05

When compared to other feature selection algorithms such as SQR, SRR, SPSOQR, and SPSORR, the rate of selection by the proposed algorithms is the lowest possible. Out of 108 features, the STRSFFQR algorithm has chosen 40 features (37 percent), the SSUSQR algorithm has chosen 54 features (50 percent), and the MDSSA algorithm has chosen 59 features (54 percent).

V. CONCLUSION

The STRSFFQR method that was suggested has provided the lowest RMSE value, which is defined as an average that is lower than 0.25 across all classification algorithms. In the pictures of acute lymphoblastic leukemia, a method known as the Improved Dominance Soft Set-based Decision Rules with Pruning (IDSSDRP) algorithm was used in order to differentiate between the blast (leukemia) cells and the non-blast (normal) cells. Based on the findings of the experiments, it can be concluded that the overall classification accuracy of the proposed IDSSDRP is as follows: 98.08 percent for the GLCM 0 dataset; 97.12 percent for the GLCM 45 dataset; 99.04 percent for the GLCM 90 dataset; 97.60 percent for the GLCM 135 dataset; and 95.67 percent for the shape and color datasets respectively. The ROC curve produced by the IDSSDRP method may be found towards the top left edge of the ROC graph, which gets increasingly relevant as it moves down. This indicates that the method under consideration is capable of accurately differentiating blast cells from non-blast cells.

REFERENCES

- [1] Begum, Shahida, Reshma Banu, Ali Ahamed, and B. D. Parameshachari. "A comparative study on improving the performance of solar power plants through IOT and predictive data analytics." In *2016 International Conference on Electrical, Electronics,*

- [2] L. Putzu, G. Caocci, and C. D. Ruberto, "Leucocyte classification for leukaemia detection using image processing techniques," *Artificial intelligence in medicine*, vol. 62, no. 3, pp. 179–191, 2014.
- [3] P. Jagadev and H. G. Virani, "Detection of leukemia and its types using image processing and machine learning," 2017 International Conference on Trends in Electronics and Informatics (ICEI), pp. 522–526, 2017.
- [4] C. Raje and J. Rangole, "Detection of Leukemia in microscopic images using image processing," 2014 International Conference on Communication and Signal Processing, pp. 255–259, 2014.
- [5] R. G. Bagasjvara, I. Candradewi, S. Hartati, and A. Harjoko, "Automated detection and classification techniques of Acute leukemia using image processing: A review," 2016 2nd International Conference on Science and Technology-Computer, pp. 35–43, 2016.
- [6] T. Hazra, M. Kumar, and D. S. S. Tripathy, "Automatic leukemia detection using image processing technique," *International Journal of Latest Technology in Engineering*, vol. 6, no. 4, pp. 42–45, 2017.
- [7] V. Shankar, M. M. Deshpande, N. Chaitra, and S. Aditi, "Automatic detection of acute lymphoblastic leukemia using image processing," 2016 IEEE International Conference on Advances in Computer Applications (ICACA), pp. 186–189, 2016.
- [8] R. D. Labati, V. Piuri, and F. Scotti, "All-IDB: The acute lymphoblastic leukemia image database for image processing," 18th IEEE international conference on image processing, pp. 2045–2048, 2011.
- [9] H. Parvaresh, H. Sajedi, and S. A. Rahimi, "Leukemia diagnosis using image processing and computational intelligence," 2018 IEEE 22nd International Conference on Intelligent Engineering Systems (INES), pp. 305–000 310, 2018.
- [10] Yang, X. S. (2010). *Nature-inspired metaheuristic algorithms*. Luniver press.
- [11] Yang, X. S., He, X. (2013). Firefly algorithm: recent advances and applications. *International Journal of Swarm Intelligence*, 1(1), 36-50.
- [12] Yüksel, S., Güzel Ergül, Z., Tozlu, N. (2014). Soft covering based rough sets and their application. *The Scientific World Journal*.
- [13] Yüksel, S., Tozlu, N., Dizman, T. H. (2015). An application of multicriteria group decision making by soft covering based rough sets. *Filomat*, 29(1), 209-219.
- [14] S. Rajpurohit, S. Patil, N. Choudhary, S. Gavasane, and P. Kosamkar, "Identification of acute lymphoblastic leukemia in microscopic blood image using image processing and machine learning algorithms," 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), pp. 2359–2363, 2018.
- [15] Parameshchari, B. D., and H. T. Panduranga. "Medical image encryption using SCAN technique and chaotic tent map system." In *Recent Advances in Artificial Intelligence and Data Engineering*, pp. 181-193. Springer, Singapore, 2022.
- [16] A. Ratley, J. Minj, and P. Patre, "Leukemia disease detection and classification using machine learning approaches: a review," 2020 First International Conference on Power, Control and Computing Technologies (ICPC2T), pp. 161–165, 2020.
- [17] M. O. Aftab, M. J. Awan, S. Khalid, R. Javed, and H. Shabir, "Executing spark BigDL for leukemia detection from microscopic images using transfer learning," 2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA), pp. 216–220, 2021.
- [18] R. Agrawal, S. Satapathy, G. Bagla, and K. Rajakumar, "Detection of white blood cell cancer using image processing," 2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN), pp. 1–6, 2019.
- [19] O. Selçuk and F. Özen, "Acute lymphoblastic leukemia diagnosis using image processing techniques," 2015 23rd Signal Processing and Communications Applications Conference (SIU), pp. 803–806, 2015.
- [20] S. Mohapatra and D. Patra, "Automated leukemia detection using hausdorff dimension in blood microscopic images," INTERACT-2010, pp. 64–68, 2010.